

Project Overview: Decision Trees, Bagging, Random Forest, and SVM Classification

Objective

This project applies both regression and classification models to real-world datasets using tree-based methods (decision trees, bagging, random forests) and Support Vector Machines (SVM). The project consists of two parts:

- Part 1: Predicting sales using regression trees and ensemble learning
- Part 2: Classifying customer purchase behavior using SVM with different kernels

Part 1: Regression Trees and Ensemble Methods (Carseats Dataset)

Objective:

Predict product Sales using the Carseats dataset with decision trees and ensemble techniques.

Workflow and Results:

a) Data Splitting

- Split into 70% training and 30% test data using a fixed seed.

b) Decision Tree Model

- Trained using the `tree()` function.
- Achieved a test MSE of 6.28.
- Visualization shows decision splits primarily on Price, CompPrice, and Advertising.

c) Cross-Validation and Pruning

- Used `cv.tree()` to identify optimal tree size.
- Pruned tree reduced overfitting and improved interpretability.
- Pruned tree MSE = 7.65, indicating slightly worse test performance than unpruned in this case.

d) Bagging Model

- Trained using `randomForest()` with all predictors (`mtry = p`).
- Achieved lowest test MSE of 2.24.
- Top predictors by importance:
 - Price
 - Shelf Location
 - CompPrice
 - Advertising

e) Random Forest

- Introduced randomness by selecting a subset of predictors (`mtry = sqrt(p)`).
- Test MSE = 3.10 (still strong, slightly higher than bagging).
- Variable importance remained similar to bagging.
- Explored effect of `mtry`:
 - Test error decreased initially with lower `mtry`, confirming reduced correlation among trees enhances generalization.

Part 2: Support Vector Machines (OJ Dataset)

Objective:

Classify whether a customer purchases CH or MM orange juice brands.

Workflow and Results:

a) Data Preparation

- Used 800 training and 270 test samples.
- Converted target variable Purchase to a factor.

b) Linear SVM

- Initial model with `cost = 0.01`: test error = 16.3%
- Hyperparameter tuning (`cost` from 0.01 to 10) found optimal `cost = 1.12`.

c) Optimal Linear SVM

- Final model error:
 - Train error = 16.8%

- Test error = 16.7%
- Performance improved slightly after tuning.

d) Radial Basis SVM

- Tuned for cost; best model had:
 - Train error = 14.8%
 - Test error = 15.2%
- Best-performing model overall due to lowest test error.

e) Polynomial Kernel SVM (Degree = 2)

- Train error = 14.5%
- Test error = 17.0%
- Slight overfitting observed (train error low, test error higher).

f) Model Comparison

Model	Training Error	Test Error
Linear SVM	16.8%	16.7%
Radial SVM	14.8%	15.2%
Polynomial SVM	14.5%	17.0%

Conclusion:

The Radial Kernel SVM outperforms other models, balancing bias and variance effectively. It achieves the lowest test error while maintaining generalization.

Summary of Insights

Decision trees are fast and interpretable but prone to overfitting. Pruning helps reduce this risk, though it may lead to underfitting if overdone. Bagging improves accuracy by reducing variance, at the cost of interpretability. Random forests build on bagging by adding randomness, offering strong generalization. Among SVM models, the radial kernel performed best, balancing accuracy and generalization though it requires careful tuning.